XRPL Validator — Daily Engineering Report · 2026-09-17

Period: 2026-09-17 00:00 to 24:00 US Eastern (2026-09-17 04:00 to 2026-09-18 04:00 UTC); ledgers #107,038,923 to #107,061,364. Every count from the ledger and the validator's logs stops at #107,061,364. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** 80 engine receipts recorded (14 fixed, 0 explained, 66 without a recorded fix).
- **Changes:** 6 deployments (cycles 135, 136, 137, 138, 139, 140), 27 commits merged into main, 2 engine findings fixed (F291–F292).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

Ledgers with a receipt in the period (#107,038,923–#107,061,364), by step:

Step	Ledgers with a receipt
Step 1 · native engine	80
Step 2 · records	0

Step	Ledgers with a receipt
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#107,044,846	Step 1 · native engine	1 transaction result(s) differed from rippled's; 16 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F291, commit 5422429, deployed in cycle 135 (2026-09-17 15:08 UTC)
R-02	—	#107,044,848	Step 1 · native engine	1 transaction result(s) differed from rippled's; 16 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F291, commit 5422429, deployed in cycle 135 (2026-09-17 15:08 UTC)
R-03	—	#107,044,849	Step 1 · native engine	1 transaction result(s) differed from rippled's; 16 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F291, commit 5422429, deployed in cycle 135 (2026-09-17 15:08 UTC)
R-04	—	#107,044,851	Step 1 · native engine	1 transaction result(s) differed from rippled's; 19 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F291, commit 5422429, deployed in cycle 135 (2026-09-17 15:08 UTC)
R-05	—	#107,044,852	Step 1 · native engine	1 transaction result(s) differed from rippled's; 19 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F291, commit 5422429, deployed in cycle 135 (2026-09-17 15:08 UTC)
R-06	—	#107,044,853	Step 1 · native engine	1 transaction result(s) differed from rippled's; 16 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F291, commit 5422429, deployed in cycle 135 (2026-09-17 15:08 UTC)
R-07	—	#107,044,854	Step 1 · native engine	1 transaction result(s) differed from rippled's; 16 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F291, commit 5422429, deployed in cycle 135 (2026-09-17 15:08 UTC)
R-08	—	#107,044,855	Step 1 · native engine	1 transaction result(s) differed from rippled's; 16 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F291, commit 5422429, deployed in cycle 135 (2026-09-17 15:08 UTC)
R-09	—	#107,044,856	Step 1 · native engine	1 transaction result(s) differed from rippled's; 16 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F291, commit 5422429, deployed in cycle 135 (2026-09-17 15:08 UTC)
R-10	—	#107,044,857	Step 1 · native engine	1 transaction result(s) differed from rippled's; 14 change(s) of ours the network did not make; 3 object(s) differed in content	fixed: F291, commit 5422429, deployed in cycle 135 (2026-09-17 15:08 UTC)
R-11	—	#107,046,318	Step 1 · native	5 change(s) of ours the network did not make; 4 object(s) differed in content	no disposition recorded

ID	Time	Ledger(s)	Step	Description	Disposition
			engine		
R-12	—	#107,048,838	Step 1 · native engine	412 change(s) of ours the network did not make	no disposition recorded
R-13	—	#107,048,839	Step 1 · native engine	1 ledger change(s) the network made were absent from ours: E6CA6BF4 (deleted); 2 change(s) of ours the network did not make; 27 object(s) differed in content	no disposition recorded
R-14	—	#107,048,840	Step 1 · native engine	1 transaction result(s) differed from rippled's; 3 ledger change(s) the network made were absent from ours: 25BFDCFF, 17B8F4C2, 8EB322D4; 25 object(s) differed in content	no disposition recorded
R-15	—	#107,048,841	Step 1 · native engine	33 transaction result(s) differed from rippled's; 4 ledger change(s) the network made were absent from ours: 19162C6F, 7E7766A2, 02FE963E; 2 change(s) of ours the network did not make; 69 object(s) differed in content	no disposition recorded
R-16	—	#107,048,842	Step 1 · native engine	3 object(s) differed in content	no disposition recorded
R-17	—	#107,048,843	Step 1 · native engine	19 object(s) differed in content	no disposition recorded
R-18	—	#107,048,844	Step 1 · native engine	1 ledger change(s) the network made were absent from ours: B335DFD1; 2 object(s) differed in content	no disposition recorded
R-19	—	#107,048,845	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-20	—	#107,048,846	Step 1 · native engine	2 ledger change(s) the network made were absent from ours: ED9B4FD5, 57103BFC; 16 object(s) differed in content	no disposition recorded
R-21	—	#107,048,847	Step 1 · native engine	4 object(s) differed in content	no disposition recorded
R-22	—	#107,048,848	Step 1 · native engine	5 change(s) of ours the network did not make; 9 object(s) differed in content	no disposition recorded
R-23	—	#107,048,849	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 ledger change(s) the network made were absent from ours: ED9B4FD5; 7 change(s) of ours the network did not make; 15 object(s) differed in content	no disposition recorded
R-24	—	#107,048,850	Step 1 · native engine	1 transaction result(s) differed from rippled's; 8 change(s) of ours the network did not make; 3 object(s) differed in content	no disposition recorded
R-25	—	#107,048,851	Step 1 · native engine	1 transaction result(s) differed from rippled's; 8 change(s) of ours the network did not make; 6 object(s) differed in content	no disposition recorded
R-26	—	#107,048,852	Step 1 · native engine	1 ledger change(s) the network made were absent from ours: ED9B4FD5; 6 change(s) of ours the network did not make; 15 object(s) differed in content	no disposition recorded
R-27	—	#107,048,853	Step 1 · native engine	2 change(s) of ours the network did not make; 2 object(s) differed in content	no disposition recorded
R-28	—	#107,048,854	Step 1 · native engine	2 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-29	—	#107,048,855	Step 1 · native	3 ledger change(s) the network made were absent from ours: 807D3FC4, ED9B4FD5 (deleted), 57103BFC; 7 change(s) of ours	no disposition recorded

ID	Time	Ledger(s)	Step	Description	Disposition
			engine	the network did not make; 15 object(s) differed in content	
R-30	—	#107,048,856	Step 1 · native engine	7 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-31	—	#107,048,857	Step 1 · native engine	4 transaction result(s) differed from rippled's; 13 ledger change(s) the network made were absent from ours: 3E23A0B3, B30B62BD, 1699EAF7; 3 change(s) of ours the network did not make; 10 object(s) differed in content	no disposition recorded
R-32	—	#107,048,858	Step 1 · native engine	2 transaction result(s) differed from rippled's; 3 ledger change(s) the network made were absent from ours: 807D3FC4, 31D501D9, 0A62337D; 11 change(s) of ours the network did not make; 22 object(s) differed in content	no disposition recorded
R-33	—	#107,048,859	Step 1 · native engine	2 transaction result(s) differed from rippled's; 2 ledger change(s) the network made were absent from ours: 269420DF, AFD4FFD4; 7 change(s) of ours the network did not make; 4 object(s) differed in content	no disposition recorded
R-34	—	#107,048,860	Step 1 · native engine	2 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-35	—	#107,048,861	Step 1 · native engine	2 transaction result(s) differed from rippled's; 5 ledger change(s) the network made were absent from ours: 416356FD, 9D19D8B9, 018B9D1A; 9 change(s) of ours the network did not make; 15 object(s) differed in content	no disposition recorded
R-36	—	#107,048,862	Step 1 · native engine	6 change(s) of ours the network did not make; 3 object(s) differed in content	no disposition recorded
R-37	—	#107,048,863	Step 1 · native engine	3 change(s) of ours the network did not make; 6 object(s) differed in content	no disposition recorded
R-38	—	#107,048,864	Step 1 · native engine	2 ledger change(s) the network made were absent from ours: 807D3FC4 (deleted), 57103BFC; 10 change(s) of ours the network did not make; 14 object(s) differed in content	no disposition recorded
R-39	—	#107,048,865	Step 1 · native engine	4 change(s) of ours the network did not make; 2 object(s) differed in content	no disposition recorded
R-40	—	#107,048,866	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 ledger change(s) the network made were absent from ours: BE04A226, 8D3CD8AB	no disposition recorded

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-17-01, INC-01 is INC-2026-09-17-01. Times are not shown where the reference node no longer holds the ledger.

40 more receipts (#107,048,867–#107,060,755) are in the record and not listed individually; their dispositions: 4 fixed, 36 no disposition recorded.

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
135	`36dbd07`	15:08:23 UTC	n/a	n/a	413 matched, 0 mismatched	#11 from 15:08 UTC: ended early: superseded by the next deployment (2026-09-17 20:01 UTC)
136	`1cb22ad`	20:01:41 UTC	n/a	n/a	407 matched, 0 mismatched	#12 from 20:01 UTC: ended early: superseded by the next deployment (2026-09-17 21:58 UTC)
137	`cec2d5a`	21:58:54 UTC	n/a	n/a	400 matched, 0 mismatched	#13 from 21:58 UTC: ended early: superseded by the next deployment (2026-09-17 23:43 UTC)
138	`45e8803`	23:43:03 UTC	n/a	n/a	416 matched, 0 mismatched	#14 from 23:43 UTC: ended early: superseded by the next deployment (2026-09-18 01:24 UTC)
139	`1e5e7b8`	01:24:46 UTC	n/a	n/a	406 matched, 0 mismatched	#15 from 01:24 UTC: ended early: superseded by the next deployment (2026-09-18 03:02 UTC)
140	`13d08f1`	03:02:51 UTC	n/a	n/a	409 matched, 0 mismatched	#16 from 03:02 UTC: ended early: superseded by the next deployment (2026-09-18 05:05 UTC)

Cycle 135 delivered: `5422429` fix(xrpl-ledger): F291 — a crossing's remainder is re-derived from the sorted fold before `done` is believed; `7bd0c74` fix(xrpl-ledger): F292 — under FlowSortStrands a strand behind the iteration's winner is not flowed, so it reaps nothing; `36dbd07` docs: Batch leg B implementation plan (native BatchTransactor, shadow attribution, devnet vectors)

Cycle 136 delivered: `46825c1` feat(xrpl-ledger): TxFields::from_json, inner_batch flag and the Batch tem codes; `6b5a309` feat(xrpl-ledger): the fee-zero preflight gate is fee_missing(), waived for batch inners; `99c4e14` feat(xrpl-ledger): apply_on_sandbox — the tx pipeline on an existing sandbox with rollback; `f5f5ab6` test(xrpl-ledger): apply_on_sandbox — exercise the do_apply rollback branch; drop an unused test import; `20af7bd` feat(xrpl-ledger): BatchTransactor — preflight, signer set, fee formula, and the four-mode inner loop; `05be0f5` fix(xrpl-ledger): BatchTransactor — rippled 3.3.0 result codes, single-inner rejection, signer cap, cleared inner results; `bd7b2c3` feat(xrpl-node): the native shadow skips Batch inner entries and attributes them to their outer; `73097b1` test(xrpl-node): devnet Batch vectors — merged outer+inner bundles, and the merge script; `07e7a11` fix(xrpl-ledger): BatchTransactor decodes base58 and hex accounts alike (decode20); `4c116ab` feat(xrpl-ledger): per-inner threading for Batch — touched-key tracking and stamp_batch_threading; `a57a648` test(xrpl-node): batch_vector — byte-exact devnet Batch vectors through the native transactor (with per-inner threading in the harness and the shadow); `c13f4c8` fix(xrpl-node): batch_vector harness warning; threadOwners coverage and a fallback receipt for the batch stamp; `90453a6` feat(xrpl-node): Batch attribution and per-inner threading in the replay tools' native legs; `e5fbfb0` fix(xrpl-node): differential_probe — guard the inner-result zip and print inner_ter in the text report; `c57e2d4` docs: Batch spec status — leg B implemented and verified on devnet; `2799e6d` fix(xrpl-ledger): Batch — fresh deferred-credits table per inner; kDisabledTxTypes; fee-formula and initiator notes; `b52cb8e` fix(xrpl-node): Batch inner pairing by id across the shadow and replay tools; inner TER counter; orphan const; `1cb22ad` docs: Batch spec — kDisabledTxTypes correction and honest status; merge script docstring

Cycle 137 delivered: `9934305` fix(ledger): OracleSet create writes Flags: 0 (finding 293); `cec2d5a` fix(ledger): own offers behind an issuer hop keep their owner credit; full delivery rests the line on pre + Amount (findings 294, 295)

Cycle 138 delivered: `45e8803` fix(ledger): reverse-pass dead-offer reaps survive the sizing snapshot (finding 296)

Cycle 139 delivered: `1e5e7b8` fix(ledger): the pool stands aside on its RAW quality, not the fee-inclusive spot (finding 297)

Cycle 140 delivered: `13d08f1` build(ffi): port the libxrpl shim to xrpld 3.4.0

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
12:31 UTC	xrpl-validator	`5422429`	fix(xrpl-ledger): F291 — a crossing's remainder is re-derived from the sorted fold before `done` is believed
12:31 UTC	xrpl-validator	`7bd0c74`	fix(xrpl-ledger): F292 — under FlowSortStrands a strand behind the iteration's winner is not flowed, so it reaps nothing
12:45 UTC	xrpl-validator	`36dbd07`	docs: Batch leg B implementation plan (native BatchTransactor, shadow attribution, devnet vectors)
18:39 UTC	xrpl-validator	`46825c1`	feat(xrpl-ledger): TxFields::from_json, inner_batch flag and the Batch tem codes
18:39 UTC	xrpl-validator	`6b5a309`	feat(xrpl-ledger): the fee-zero preflight gate is fee_missing(), waived for batch inners
18:39 UTC	xrpl-validator	`99c4e14`	feat(xrpl-ledger): apply_on_sandbox — the tx pipeline on an existing sandbox with rollback
18:39 UTC	xrpl-validator	`f5f5ab6`	test(xrpl-ledger): apply_on_sandbox — exercise the do_apply rollback branch; drop an unused test import
18:39 UTC	xrpl-validator	`20af7bd`	feat(xrpl-ledger): BatchTransactor — preflight, signer set, fee formula, and the four-mode inner loop
18:39 UTC	xrpl-validator	`05be0f5`	fix(xrpl-ledger): BatchTransactor — rippled 3.3.0 result codes, single-inner rejection, signer cap, cleared inner results
18:39 UTC	xrpl-validator	`bd7b2c3`	feat(xrpl-node): the native shadow skips Batch inner entries and attributes them to their outer
18:39 UTC	xrpl-validator	`73097b1`	test(xrpl-node): devnet Batch vectors — merged outer+inner bundles, and the merge script
18:39 UTC	xrpl-validator	`07e7a11`	fix(xrpl-ledger): BatchTransactor decodes base58 and hex accounts alike (decode20)
18:39 UTC	xrpl-validator	`4c116ab`	feat(xrpl-ledger): per-inner threading for Batch — touched-key tracking and stamp_batch_threading
18:39 UTC	xrpl-validator	`a57a648`	test(xrpl-node): batch_vector — byte-exact devnet Batch vectors through the native transactor (with per-inner threading in the harness and the shadow)
18:39 UTC	xrpl-validator	`c13f4c8`	fix(xrpl-node): batch_vector harness warning; threadOwners coverage and a fallback receipt for the batch stamp
18:39 UTC	xrpl-validator	`90453a6`	feat(xrpl-node): Batch attribution and per-inner threading in the replay tools' native legs
18:39 UTC	xrpl-validator	`e5fbfb0`	fix(xrpl-node): differential_probe — guard the inner-result zip and print inner_ter in the text report
18:39 UTC	xrpl-validator	`c57e2d4`	docs: Batch spec status — leg B implemented and verified on devnet
18:39 UTC	xrpl-validator	`2799e6d`	fix(xrpl-ledger): Batch — fresh deferred-credits table per inner; kDisabledTxTypes; fee-formula and initiator notes
18:39 UTC	xrpl-validator	`b52cb8e`	fix(xrpl-node): Batch inner pairing by id across the shadow and replay tools; inner TER counter; orphan const

Merged	Repo	Commit	Subject
18:39 UTC	xrpl-validator	`1cb22ad`	docs: Batch spec — kDisabledTxTypes correction and honest status; merge script docstring
20:41 UTC	xrpl-validator	`9934305`	fix(ledger): OracleSet create writes Flags: 0 (finding 293)
20:41 UTC	xrpl-validator	`cec2d5a`	fix(ledger): own offers behind an issuer hop keep their owner credit; full delivery rests the line on pre + Amount (findings 294, 295)
22:22 UTC	xrpl-validator	`45e8803`	fix(ledger): reverse-pass dead-offer reaps survive the sizing snapshot (finding 296)
00:13 UTC	xrpl-validator	`1e5e7b8`	fix(ledger): the pool stands aside on its RAW quality, not the fee-inclusive spot (finding 297)
01:54 UTC	xrpl-validator	`13d08f1`	build(ffi): port the libxrpl shim to xrpld 3.4.0
03:47 UTC	xrpl-validator	`a1aecb0`	fix(ledger): a multi-path pool is gated by its own book's tip, not only the rival strand (finding 298)

Findings fixed

- **fix(xrpl-ledger): F291 — a crossing's remainder is re-derived from the sorted fold before `done` is believed** (`5422429`, xrpl-validator) rippled recomputes remainingOut at every iteration boundary as `outReq - sum(savedOuts)` over the ASCENDING multiset (StrandFlow.h:791); the running chain is never consulted. Ten consecutive tecKILLED receipts (107044846-857, rGH4WSUU's Fill-or-Kill RLUSD buys) had four iterations whose chain decrements to exactly zero while the sorted 16-digit fold is 45.66475799999999 against 45.664758 wanted: a 1e-14 crumb. …
- **fix(xrpl-ledger): F292 — under FlowSortStrands a strand behind the iteration's winner is not flowed, so it reaps nothing** (`7bd0c74`, xrpl-validator) rippled flows an iteration's admitted strands in bound order and breaks at the first that succeeds and passes limitQuality (StrandFlow.h: `best.emplace(...); pushRemainingCurToNext(strandIndex + 1); break;`). The strands behind the winner are not flowed that iteration: no rev pass, no offer stream, nothing reaped.

Other engineering changes

- **feat(xrpl-ledger): TxFields::from_json, inner_batch flag and the Batch tem codes** (`46825c1`, xrpl-validator) Add `TxFields::inner_batch: bool` plus `fee_missing()` (waived for batch inners) and `from_json()`, moved from xrpl-node's native_apply::build_txfields (now a one-line delegate). from_json decodes the Account field via tx::offer::decode20, which accepts the engine's hex account dialect as well as base58 — build_txfields's original decode_address call was base58-only and could not read the hex Account strings the …
- **feat(xrpl-ledger): the fee-zero preflight gate is fee_missing(), waived for batch inners** (`6b5a309`, xrpl-validator)
- **feat(xrpl-ledger): apply_on_sandbox — the tx pipeline on an existing sandbox with rollback** (`99c4e14`, xrpl-validator) Mirrors xrpl_node::native_apply::native_apply_one but runs on a caller-supplied Sandbox instead of allocating a fresh one, restoring the entry snapshot on every non-applied path and keeping only the common changes on a claimed tec. Task 4 will call this per inner transaction of a Batch.
- **feat(xrpl-ledger): BatchTransactor — preflight, signer set, fee formula, and the four-mode inner loop** (`20af7bd`, xrpl-validator)

- **fix(xrpl-ledger): BatchTransactor — rippled 3.3.0 result codes, single-inner rejection, signer cap, cleared inner results** (`05be0f5`, xrpl-validator)
- **feat(xrpl-node): the native shadow skips Batch inner entries and attributes them to their outer** (`bd7b2c3`, xrpl-validator) rippled records every Batch (BatchV1_1) inner transaction as its own ledger entry — own hash, own metadata carrying ParentBatchID, the indices right after its outer — but APPLIES it inside the outer's application. Our BatchTransactor::do_apply does the same, so the shadow's replay must skip the inner entries and compare the outer's mutation set against the union of the outer's and its inners' AffectedNodes.
- **fix(xrpl-ledger): BatchTransactor decodes base58 and hex accounts alike (decode20)** (`07e7a11`, xrpl-validator) `decode_account` accepted only the 40-hex-char dialect, on the assumption that `native_apply::hexify_addresses` had already rewritten every r-address in the transaction. It has not: that pass runs on ledger-state pre-images, and `TxFields::from_json` only rewrites the outer's own top-level account fields — nested `RawTransaction` / `BatchSigner` objects keep whatever dialect the feed sent. Every real Batch therefore …
- **feat(xrpl-ledger): per-inner threading for Batch — touched-key tracking and stamp_batch_threading** (`4c116ab`, xrpl-validator) rippled applies every Batch inner as its own transaction (`applyBatchTransactions -> apply(..., inner, tapBATCH)`), so each ledger object an inner touched threads the INNER's transaction id, while the outer's own changes (its fee and sequence) thread the outer's. The last toucher wins. Our engine runs the inners inside the outer's `do_apply` and the callers stamped the whole mutation set with the outer's hash, …
- **fix(xrpl-node): batch_vector harness warning; threadOwners coverage and a fallback receipt for the batch stamp** (`c13f4c8`, xrpl-validator) `assert_expect` took a `&LedgerState` it never read, which the test target reported as `unused variable: state` — the parameter is gone from the signature and both call sites (the earlier clippy sweep never linted the file: it exits on the pre-existing errors in xrpl-ledger before reaching the node's test targets).
- **feat(xrpl-node): Batch attribution and per-inner threading in the replay tools' native legs** (`90453a6`, xrpl-validator) state_replay and differential_probe iterate a ledger's entries the same way the native shadow does, and both were still applying Batch INNER entries standalone — which the engine rejects out of hand (tfInnerBatchTxn with Fee "0") — and stamping the outer's mutations with the outer's hash alone. Every Batch ledger therefore read as a divergence on every inner and a threading byte-diff on the outer.
- **fix(xrpl-node): differential_probe — guard the inner-result zip and print inner_ter in the text report** (`e5fbfb0`, xrpl-validator) The kept inner ids are the ones the FIXTURE carries; the per-inner results are the inners the engine ATTEMPTED. Pairing them by index is meaningful only while the two lists agree — one hole shifts every later inner onto its neighbour's result and reports a confident, wrong verdict. The per-index comparison is now guarded by a length check; on disagreement it withholds the inner verdicts, prints one BATCH-INNER …
- **fix(xrpl-ledger): Batch — fresh deferred-credits table per inner; kDisabledTxTypes; fee-formula and initiator notes** (`2799e6d`, xrpl-validator)
- **fix(xrpl-node): Batch inner pairing by id across the shadow and replay tools; inner TER counter; orphan const** (`b52cb8e`, xrpl-validator)
- **fix(ledger): OracleSet create writes Flags: 0 (finding 293)** (`9934305`, xrpl-validator) `sfFlags` is soeREQUIRED on the Oracle ledger format, so a created Oracle carries `Flags: 0` right after its type. The create path built the object without it and serialized five bytes short (#107052956 B1EB9B20D12E, 172 bytes against

mainnet's 177, the diff starting at byte 3) — the same shape as finding 187 on DID. New oracle_vector suite pins the object.

- **fix(ledger): own offers behind an issuer hop keep their owner credit; full delivery rests the line on pre + Amount (findings 294, 295)** (`cec2d5a`, xrpl-validator) #107052630 32386DDEB6B8: rUnRkdr pays 117.732708 USDT with FIL through the explicit issuer hop [rsL5Y] and the XRP bridge; strand 1 crosses the sender's OWN two FIL/USDT offers.
- **fix(ledger): reverse-pass dead-offer reaps survive the sizing snapshot (finding 296)** (`45e8803`, xrpl-validator) #107009438 2877CBCC88C9: a deliver-max partial payment of 166716 drops for RLUSD through USDC.axl. rippled's reverse pass, asked for everything, walks the whole USDC.axl/RLUSD book: past the one funded tip it steps over rDeXHa's three unfunded offers and marks them ofrsToRm; the forward pass buys 0.2129 RLUSD from the tip alone, and the driver still deletes the three (plus book pages, owner page, OwnerCount) after …
- **fix(ledger): the pool stands aside on its RAW quality, not the fee-inclusive spot (finding 297)** (`1e5e7b8`, xrpl-validator) #107056200 BD9C7473B84F: rhTsmUJ's 6 XRP partial self-payment into RVR meets rMBPaL7's fresh offer at the tip and a 0.506% pool. rippled's `AMMLiquidity::getOffer` (AMMLiquidity.cpp:181-183) stands the pool aside only when `Quality{balances}` — the raw pool quality, no trading fee — is not strictly better than the tip or sits within 1e-7 of it; the fee enters only in the anchored offer `changeSpotPriceQuality` …
- **fix(ledger): a multi-path pool is gated by its own book's tip, not only the rival strand (finding 298)** (`a1aecb0`, xrpl-validator) #107060755 F43C3C4DA037: rsdsSA7's tfSell offer, 852332.25 ASC for 599530.50 PLR, meets a direct ASC/PLR pool and a two-pool XRP bridge — multi-path, Fibonacci offers. rippled takes five direct-pool slices (25.645 ASC), then `AMMLiquidity::getOffer` (AMMLiquidity.cpp:181-196) stands the pool aside because its RAW quality (1.40353) no longer beats its own book's tip — the taker's own offer 4CF7660E at 1.401935, still …

Tests and campaigns

- **test(xrpl-ledger): apply_on_sandbox — exercise the do_apply rollback branch; drop an unused test import** (`f5f5ab6`, xrpl-validator)
- **test(xrpl-node): devnet Batch vectors — merged outer+inner bundles, and the merge script** (`73097b1`, xrpl-validator)
- **test(xrpl-node): batch_vector — byte-exact devnet Batch vectors through the native transactor (with per-inner threading in the harness and the shadow)** (`a57a648`, xrpl-validator)

Docs and operations

- **docs: Batch leg B implementation plan (native BatchTransactor, shadow attribution, devnet vectors)** (`36dbd07`, xrpl-validator)
- **docs: Batch spec status — leg B implemented and verified on devnet** (`c57e2d4`, xrpl-validator)
- **docs: Batch spec — kDisabledTxTypes correction and honest status; merge script docstring** (`1cb22ad`, xrpl-validator)
- **build(ffi): port the libxrpl shim to xrpld 3.4.0** (`13d08f1`, xrpl-validator)

Independence program

No independence record for UTC day 2026-09-17; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
BatchV1_1	2026-09-15 14:06 UTC	2026-09-29 14:06 UTC

Amendments enabled on mainnet: 94.

Open items

- Receipt R-11 (#107,046,318): open, under investigation.
- Receipt R-12 (#107,048,838): open, under investigation.
- Receipt R-13 (#107,048,839): open, under investigation.
- Receipt R-14 (#107,048,840): open, under investigation.
- Receipt R-15 (#107,048,841): open, under investigation.
- Receipt R-16 (#107,048,842): open, under investigation.
- Receipt R-17 (#107,048,843): open, under investigation.
- Receipt R-18 (#107,048,844): open, under investigation.
- Receipt R-19 (#107,048,845): open, under investigation.
- Receipt R-20 (#107,048,846): open, under investigation.
- Receipt R-21 (#107,048,847): open, under investigation.
- Receipt R-22 (#107,048,848): open, under investigation.
- Receipt R-23 (#107,048,849): open, under investigation.
- Receipt R-24 (#107,048,850): open, under investigation.
- Receipt R-25 (#107,048,851): open, under investigation.
- Receipt R-26 (#107,048,852): open, under investigation.
- Receipt R-27 (#107,048,853): open, under investigation.
- Receipt R-28 (#107,048,854): open, under investigation.
- Receipt R-29 (#107,048,855): open, under investigation.
- Receipt R-30 (#107,048,856): open, under investigation.

- Receipt R-31 (#107,048,857): open, under investigation.
- Receipt R-32 (#107,048,858): open, under investigation.
- Receipt R-33 (#107,048,859): open, under investigation.
- Receipt R-34 (#107,048,860): open, under investigation.
- Receipt R-35 (#107,048,861): open, under investigation.
- Receipt R-36 (#107,048,862): open, under investigation.
- Receipt R-37 (#107,048,863): open, under investigation.
- Receipt R-38 (#107,048,864): open, under investigation.
- Receipt R-39 (#107,048,865): open, under investigation.
- Receipt R-40 (#107,048,866): open, under investigation.
- Receipt R-41 (#107,048,867): open, under investigation.
- Receipt R-42 (#107,048,868): open, under investigation.
- Receipt R-43 (#107,048,869): open, under investigation.
- Receipt R-44 (#107,048,870): open, under investigation.
- Receipt R-45 (#107,048,871): open, under investigation.
- Receipt R-46 (#107,048,873): open, under investigation.
- Receipt R-47 (#107,048,875): open, under investigation.
- Receipt R-48 (#107,048,876): open, under investigation.
- Receipt R-49 (#107,048,877): open, under investigation.
- Receipt R-50 (#107,048,878): open, under investigation.
- Receipt R-51 (#107,048,879): open, under investigation.
- Receipt R-52 (#107,048,880): open, under investigation.
- Receipt R-53 (#107,048,882): open, under investigation.
- Receipt R-54 (#107,048,885): open, under investigation.
- Receipt R-55 (#107,048,886): open, under investigation.
- Receipt R-56 (#107,048,887): open, under investigation.
- Receipt R-57 (#107,048,888): open, under investigation.
- Receipt R-58 (#107,048,890): open, under investigation.
- Receipt R-59 (#107,048,892): open, under investigation.
- Receipt R-60 (#107,048,894): open, under investigation.
- Receipt R-61 (#107,048,896): open, under investigation.
- Receipt R-62 (#107,048,899): open, under investigation.
- Receipt R-63 (#107,048,901): open, under investigation.
- Receipt R-64 (#107,048,903): open, under investigation.
- Receipt R-65 (#107,048,905): open, under investigation.
- Receipt R-66 (#107,048,906): open, under investigation.
- Receipt R-67 (#107,048,907): open, under investigation.
- Receipt R-68 (#107,048,908): open, under investigation.
- Receipt R-69 (#107,048,909): open, under investigation.

- Receipt R-70 (#107,048,910): open, under investigation.
- Receipt R-71 (#107,048,911): open, under investigation.
- Receipt R-72 (#107,048,912): open, under investigation.
- Receipt R-73 (#107,048,913): open, under investigation.
- Receipt R-74 (#107,048,914): open, under investigation.
- Receipt R-75 (#107,048,916): open, under investigation.
- Receipt R-76 (#107,048,919): open, under investigation.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Permission-Gated SingleAssetVault Deposits** (compliance) Add an optional sfDomainID SField to the Vault ledger object (ltVAULT), settable only at VaultCreate, referencing an existing PermissionedDomain. When
- **Pseudo-Account Custody Sub-Ledgers** (custody) Add a SubAccountCreate transaction that mints a pseudo-account per the XLS-64 keyless-account derivation (keylet::subAccount(parentID, seq)), recorded

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-17 12:00	09-17 12:31	xrpl-validator	`5422429`	fix(xrpl-ledger): F291 — a crossing's remainder is re-derived from the sorted fold before `done` is believed
09-17 12:31	09-17 12:31	xrpl-validator	`7bd0c74`	fix(xrpl-ledger): F292 — under FlowSortStrands a strand behind the iteration's winner is not flowed, so it reaps nothing
09-17 12:45	09-17 12:45	xrpl-validator	`36dbd07`	docs: Batch leg B implementation plan (native BatchTransactor, shadow attribution, devnet vectors)
09-17 13:38	09-17 18:39	xrpl-validator	`46825c1`	feat(xrpl-ledger): TxFields::from_json, inner_batch flag and the Batch tem codes
09-17 13:48	09-17 18:39	xrpl-validator	`6b5a309`	feat(xrpl-ledger): the fee-zero preflight gate is fee_missing(), waived for batch inners
09-17 13:57	09-17 18:39	xrpl-validator	`99c4e14`	feat(xrpl-ledger): apply_on_sandbox — the tx pipeline on an existing sandbox with rollback
09-17 14:06	09-17 18:39	xrpl-validator	`f5f5ab6`	test(xrpl-ledger): apply_on_sandbox — exercise the do_apply rollback branch; drop an unused test import
09-17 14:18	09-17 18:39	xrpl-validator	`20af7bd`	feat(xrpl-ledger): BatchTransactor — preflight, signer set, fee formula, and the four-mode inner loop

Committed	Merged into main	Repo	Commit	Subject
09-17 14:40	09-17 18:39	xrpl-validator	`05be0f5`	fix(xrpl-ledger): BatchTransactor — rippled 3.3.0 result codes, single-inner rejection, signer cap, cleared inner results
09-17 15:01	09-17 18:39	xrpl-validator	`bd7b2c3`	feat(xrpl-node): the native shadow skips Batch inner entries and attributes them to their outer
09-17 15:16	09-17 18:39	xrpl-validator	`73097b1`	test(xrpl-node): devnet Batch vectors — merged outer+inner bundles, and the merge script
09-17 15:40	09-17 18:39	xrpl-validator	`07e7a11`	fix(xrpl-ledger): BatchTransactor decodes base58 and hex accounts alike (decode20)
09-17 15:49	09-17 18:39	xrpl-validator	`4c116ab`	feat(xrpl-ledger): per-inner threading for Batch — touched-key tracking and stamp_batch_threading
09-17 15:49	09-17 18:39	xrpl-validator	`a57a648`	test(xrpl-node): batch_vector — byte-exact devnet Batch vectors through the native transactor (with per-inner threading in the harness and the shadow)
09-17 16:05	09-17 18:39	xrpl-validator	`c13f4c8`	fix(xrpl-node): batch_vector harness warning; threadOwners coverage and a fallback receipt for the batch stamp
09-17 16:30	09-17 18:39	xrpl-validator	`90453a6`	feat(xrpl-node): Batch attribution and per-inner threading in the replay tools' native legs
09-17 16:42	09-17 18:39	xrpl-validator	`e5fbfb0`	fix(xrpl-node): differential_probe — guard the inner-result zip and print inner_ter in the text report
09-17 16:52	09-17 18:39	xrpl-validator	`c57e2d4`	docs: Batch spec status — leg B implemented and verified on devnet
09-17 17:20	09-17 18:39	xrpl-validator	`2799e6d`	fix(xrpl-ledger): Batch — fresh deferred-credits table per inner; kDisabledTxTypes; fee-formula and initiator notes
09-17 17:28	09-17 18:39	xrpl-validator	`b52cb8e`	fix(xrpl-node): Batch inner pairing by id across the shadow and replay tools; inner TER counter; orphan const
09-17 17:29	09-17 18:39	xrpl-validator	`1cb22ad`	docs: Batch spec — kDisabledTxTypes correction and honest status; merge script docstring
09-17 20:15	09-17 20:41	xrpl-validator	`9934305`	fix(ledger): OracleSet create writes Flags: 0 (finding 293)
09-17 20:32	09-17 20:41	xrpl-validator	`cec2d5a`	fix(ledger): own offers behind an issuer hop keep their owner credit; full delivery rests the line on pre + Amount (findings 294, 295)
09-17 22:18	09-17 22:22	xrpl-validator	`45e8803`	fix(ledger): reverse-pass dead-offer reaps survive the sizing snapshot (finding 296)
09-18 00:01	09-18 00:13	xrpl-validator	`1e5e7b8`	fix(ledger): the pool stands aside on its RAW quality, not the fee-inclusive spot (finding 297)
09-18 01:54	09-18 01:54	xrpl-validator	`13d08f1`	build(ffi): port the libxrpl shim to xrpld 3.4.0
09-18 03:47	09-18 03:47	xrpl-validator	`a1aecb0`	fix(ledger): a multi-path pool is gated by its own book's tip, not only the rival strand (finding 298)

Times in UTC. A commit appears on the day it was committed to main or merged into it.